UNC514: DATA SCIENCE FUNDAMENTALS

L	T	P	Cr
2	0	2	3.0

Course Objective:
To elaborate the basics of data science and provide a foundation for understanding the challenges and applications.

Introduction to Python: Python ecosystem (Anaconda, Jupyter, VS Code), Basic syntax, variables, Operators, Data types (Numbers, Booleans, Strings), Control Flow, Loops, Sequence Generation (range function), String Operations, Exception Handling, Command Line Arguments, Use of Libraries, File Handling (Read, Write, Merge, etc).

Data Structures in Python: List, Tuple, Sets, Dictionary, Operations on Data Structures (Declarations, Iterations, Adding/deleting element, min/max/sorting, merge, select), OOPs in Python, working with Numpy, Pandas and Scikit-Learn.

Data science Lifecycle and Workflows. Handling structured data from CSV, Excel, JSON, and SQL. Data cleaning, wrangling, preprocessing, missing value handling, normalization, and scaling. Descriptive statistics: measures of central tendency and variability. Feature engineering and selection. Data visualization principles. Plotting using Matplotlib (line, bar, histogram, box, scatter). Introduction to Seaborn and interactive visualization using Plotly.

Advances in Data Science: Basics of Correlation, Regression, Probability and Random Variables, Models for Structured and Unstructured Data, Predictive Analytics of Structured and Unstructured Data, Bias, Variance, and Overfitting Concepts, Neuro-Symbolic AI in Data Science: Introduction to combining data-driven learning with symbolic reasoning through logic-aware data representation. Rule-based feature engineering and constraint-driven data validation to embed domain knowledge into datasets.

Case Studies: Sensor Data Analysis for Environmental Monitoring, Power Consumption Analysis of Electrical and Electronic Devices, Biomedical Signal Data Analysis.

Familiarization with standard: IEEE 7000™-2021: Model Process for Addressing Ethical Concerns during System Design, IEEE 2791-2020: Standard for Genome Data Analysis Methods, IEEE 2805.2-2025: Standard for Data Acquisition, IEEE 2894-2024: Explainable AI, IEEE 3404-2025: for Sharing Data and Models for AI.

Laboratory Work
Implementation of various Python's data strictures, data wrangling, visualization, analytics techniques on real world problems using Python.

Course Learning Objectives (CLO)
The students will be able to:

1. Apply python basics, data structures, and scientific libraries to clean, transform, and prepare structured and unstructured data.
2. Explain the data science lifecycle and apply structured workflows to handle data from CSV, Excel, JSON, and SQL sources.
3. Analyze and interpret relationships, uncertainty, and trends in structured and unstructured data using correlation, regression concepts, probability measures, and predictive analytics

techniques

4. Apply and justify logic-aware and ethical data science practices by implementing rule-based feature construction, constraint-driven data validation, and explaining bias, variance, and overfitting in analytics workflows.

Text Books

1. Jiawei Han, Micheline Kamber, Jian Pei ,Data Mining Concepts and Techniques, (3rd Ed.),Morgan Kaufmann
2. Roger D. Peng R Programming for Data Science

Reference Books

1. Trevor Hastie Robert Tibshirani Jerome Friedman, The Elements of Statistical Learning, Springer